Business Memo: RTO Prevention System

To: Founder, D2C Haircare Store
From: Keerthi
Subject: Reducing Cash-on-Delivery Return-to-Origin (RTO) Losses
Date: 31 October 2026

1. Business Problem

The store receives approximately 3,000 orders per month, with 55% placed through Cash on Delivery (COD). COD orders have an approximate RTO rate of 28%, compared with 4% for prepaid orders.

Each RTO creates an estimated cost of ₹169.40, including forward shipping, return shipping, packaging and expected product damage. The objective is to identify potentially risky COD orders before dispatch and recommend appropriate actions without unnecessarily rejecting genuine customers.

2. Key Dataset Findings

The cleaned dataset contains 1,987 completed, unique orders, with an overall historical RTO rate of approximately 28.43%.

Three useful patterns were identified:

- **Geographic differences:** Bihar has an approximate RTO rate of 53%, Uttar Pradesh 45%, and Maharashtra 18%. Location can contribute to risk assessment, but entire states should not be automatically blocked.
- **Customer history:** Repeat customers with no previous RTOs have an approximate 12% failure rate, compared with around 30% for new customers. First-time customers should not automatically be treated as risky.
- **Order value:** Orders worth ₹1,000–₹1,500 have an approximate 20% failure rate, while orders above ₹1,500 have an approximate 38% failure rate. Higher-value orders may require additional verification.

These patterns are historical associations, not proof that a particular customer or location will result in an RTO.

3. Proposed Decision System

The system uses a trained Random Forest model to estimate the probability of an order becoming an RTO. Configurable thresholds convert the predicted risk into four actions.

Predicted RTO risk	Recommended action
Below 20%	Ship as-is
20% to below 50%	Nudge to prepaid

50% to below 70%	Confirm through WhatsApp
70% and above	Hold for manual review

A ₹20 prepaid discount is offered as an initial business assumption. The WhatsApp simulator supports YES, NO and no-response scenarios, with a 24-hour timeout. High-risk orders are flagged for manual review rather than automatically cancelled.

4. September Simulation and Financial Impact

The Random Forest model was applied to all 329 orders from September 2026 using the same 20%, 50% and 70% thresholds.

Decision	**Orders**	**Average predicted risk**	**Actual RTOs**
Ship as-is	28	16.87%	0
Nudge to prepaid	216	32.12%	29
WhatsApp confirmation	58	59.55%	31
Hold for manual review	27	78.27%	26
Total	**329**		**86**

The actual September RTO rate was 26.14%. The average order value for September was approximately ₹878.40, while the overall dataset average was approximately ₹919.96.

The estimated financial impact is:

Metric	**Estimated amount**
Potential RTO savings	₹3,579.93
Prepaid discount cost	₹1,296.00
Genuine-order contribution at risk	₹3,056.95
Prepaid net benefit	₹1,338.51
Estimated net impact	**₹1,861.49**

The financial simulation assumes a 30% prepaid conversion rate, a ₹20 discount, and COD and prepaid RTO rates of 28% and 4%, respectively. It assumes held orders are not shipped. These are estimates, not verified savings or measured customer conversions. The WhatsApp confirmation benefit is not separately included in this calculation.

Operations vs Marketing: Operations benefits from identifying risky orders and reducing potential RTO exposure. Marketing must balance prepaid incentives against discount costs and

the possibility of losing genuine customers. Both teams should monitor the actual results before increasing incentives or restricting orders.

The ₹300 customer acquisition cost is not deducted from the dispatch decision because it has already been incurred at that stage. Repeat purchases and customer lifetime value should instead be considered in longer-term decisions.

5. Risks and Monitoring

The model does not guarantee that every high-risk order will become an RTO. Incorrectly holding genuine orders can affect customer experience and revenue. Regional patterns may also create unfair decisions if used without human review.

The model's test performance was approximately 73.87% accuracy and 0.696 ROC-AUC. It identified fewer than half of the actual RTO cases in the test set, so predictions should support decisions rather than replace human judgement.

The business should monitor actual RTO rates, prepaid conversion, WhatsApp responses, false-positive holds, discount expenditure and net financial impact. Predictions should be compared with actual outcomes regularly.

6. Next Two Weeks

1. Connect the decision engine to incoming Shopify orders through a webhook.
2. Integrate the WhatsApp Business API for real confirmation messages and prepaid payment links.
3. Store customer responses and final delivery outcomes for future analysis.
4. Build an operations review queue for high-risk and incomplete-address orders.
5. Compare prepaid discount strategies and measure actual conversion.
6. Evaluate model performance using new order data and monitor false-positive decisions.

Conclusion

The proposed system combines machine-learning risk prediction, configurable business rules, WhatsApp communication and a founder dashboard. It aims to reduce avoidable RTO costs while protecting genuine customer orders through manual review and continuous monitoring.